SuppMat_VVunitAligner

User Manual — SuppMat_VVunitAligner.praat

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https://github.com/leonidasjr/VVunitAlignerCode_webMAUS

Pre-processing step for SpeechRhythmExtractor.praat | Supplementary Material

0. About this document

This manual accompanies SuppMat_VVunitAligner.praat. It is addressed to users with no prior programming experience. Every section of the 717-line script is documented below, aligned to the exact line numbers of the file in the repository. This script must be run before SpeechRhythmExtractor.praat.

1. What the script does

SuppMat_VVunitAligner.praat is a Praat pre-processing script. It takes the output of the webMAUS forced-aligner (Kisler et al., 2017) — the TextGrid produced by the G2P→MAUS→PHO2SYL pipeline — and re-aligns the canonical/phonological syllable units (PHO2SYL) into phonetic V-to-V (vowel-onset-to-vowel-onset, VV) units. These VV units are the syllable-sized intervals required by SpeechRhythmExtractor.praat as V_to_V_tier = 1.

For each audio file the script produces three TextGrid objects in the Praat Objects window:

- MAUS_[filename]: the original webMAUS segmentation (preserved unchanged)
- Complete_[filename]: the full 10-tier blend (V/C/#, VV, phonological syllables, words, chunks, tone marks) used by SpeechRhythmExtractor
- The [filename].TextGrid working object to be saved for later being used by SpeechRhythmExtractor

1.1 The webMAUS Italian (IT) workaround for Brazilian Portuguese

webMAUS does not offer a dedicated Brazilian Portuguese (BP) phoneme inventory. The Italian (IT) protocol produces the best approximation for BP phone-level forced alignment Bisol, 2010; Callou; Leite, 1990; Canepay, 1999; Major, 2001). When processing BP data, always select Italian (IT) in webMAUS and Portuguese (BR) in this script. Manual correction of the VC tier is strongly recommended before running SpeechRhythmExtractor (cf. Barbosa and Madureira, 2015; Ladefoged, 2003; for protocol of manual correction).

2. Complete workflow

The script is the middle step of a three-stage pipeline:

1. STAGE 1 — webMAUS: upload filename.wav + filename.txt → Pipeline: G2P→MAUS→PHO2SYL → Language: English (US) or Italian (IT) for BP → Output: Praat (TextGrid) → Keep everything: false → Download ZIP → extract filename.TextGrid
2. STAGE 2 — VVUnitAligner (this script): place script + filename.wav + filename.TextGrid in the same folder → click on “Save Textgrid files” → run scriptt in Praat

3. STAGE 3 — SpeechRhythmExtractor: use Complete_[filename].TextGrid as input, setting V_to_V_tier = 1, V_C_Pause_tier = 2, Word_tier = 4, Chunk_tier = 5 (mapped to tier 5 by SpeechRhythmExtractor)

3. Requirements and folder structure

3.1 Software

Praat 5 or higher — https://www.praat.org. (free)

3.2 Folder structure

All files must be in the same folder as the script:

```
project_folder/
  ├── SuppMat_VVunitAligner.praat
  ├── PBAFEM001.wav
  ├── PBAFEM001.TextGrid (from webMAUS)
  └── ...
```

> ℹ *The .wav and .TextGrid filenames must be identical. Keep the folder path short (e.g., Desktop) to avoid long-path errors.*

4. webMAUS settings (Stage 1)

Use exactly these settings at:

https://clarin.phonetik.uni-muenchen.de/BASWebServices/interface/Pipeline:

Setting	Value / action
Service	Pipeline without ASR
URL	https://clarin.phonetik.uni-muenchen.de/BASWebServices/interface/Pipeline
Pipeline name	G2P → MAUS → PHO2SYL
Language — English data	English (US)
Language — Portuguese (BR) data	Italian (IT) | ⚠ No BP protocol available in webMAUS
Output format	Praat (TextGrid)
Keep everything	false
Required files	filename.wav + filename.txt (identical base names; no spaces)
Download	'Download as ZIP file' → extract → place .TextGrid in script folder

> ℹ *For Brazilian Portuguese data, always select Italian (IT) as the Language in webMAUS. This is a known limitation of the BAS Web Services. The tests are supported by previous literature (cf. Bisol, 2010; Callou; Leite, 1990; Canepay, 1999; Major, 2001 among others). Manual correction of the resulting VC tier is strongly recommended (Barbosa & Madureira, 2015; Ladefoged, 2003).*

5. Running the script (Stage 2)

5.1 Opening and running

4. Open Praat.
5. Click Praat → Open Praat script…
6. Select SuppMat_VVunitAligner.praat.
7. Script Editor opens. Click Run → Run (Ctrl+R / Cmd+R).

5.2 The input form (lines 12–30)

The dialog shown in Figure 1 will appear. Line numbers in parentheses refer to the script.

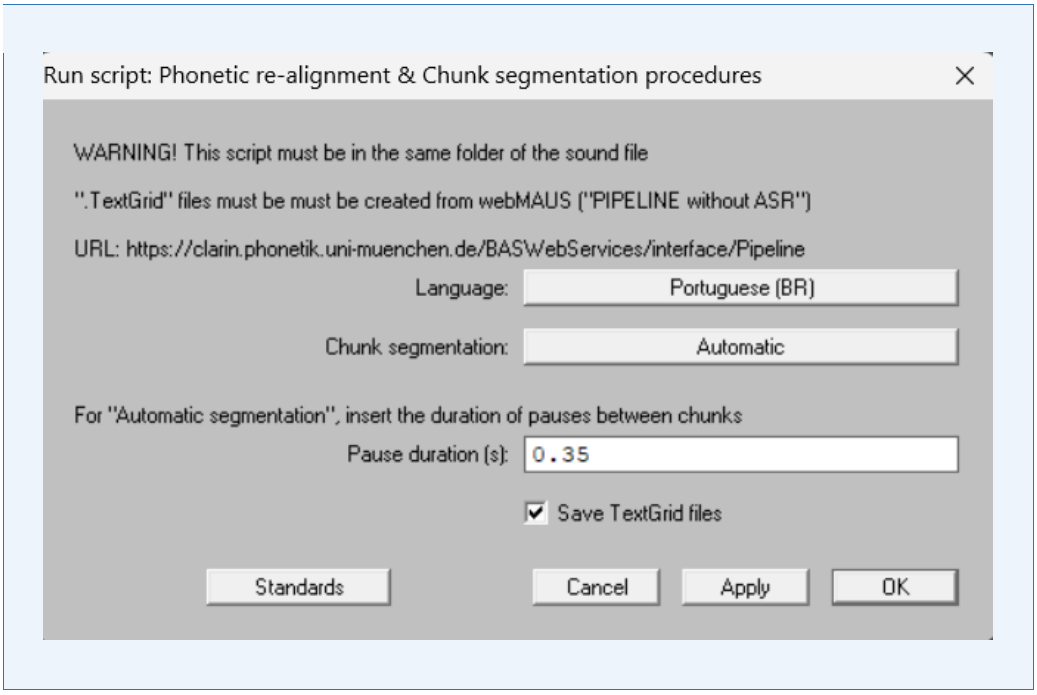

Figure 1. The SuppMat_VVunitAligner input form. Insert a screenshot of the Praat dialog here.

Field (line)	Default	Type	Description
`Language (17)`	`1 = PT(BR)`	optionmenu	1 = Portuguese (BR) — use Italian (IT) in webMAUS. 2 = English (US).
`Chunk_segmentation`[1] `(23)`	`1 = Automatic`	optionmenu	1 = Automatic (intensity); 2 = Forced/manual (interactive); 3 = None.
`Pause_duration_(s) (28)`	`0.35`	positive	Min. silence gap (s) for Automatic chunk boundaries.

[1] Automatic segmentation was performed using forced alignment (webMAUS), followed by automatic re-alignment (Figure 1) based on intensity (50 dB) and minimum pause duration (350 ms) thresholds. Forced alignment provides initial segmental boundaries (Kisler; Reichel; Schiel, 2017; Lee, 2012), while acoustic-temporal criteria refine these boundaries and reduce signal-related inaccuracies (Yuan et al., 2023). The selection of these thresholds is aligned with Voice Activity Detection (VAD) research, where acoustic energy and duration are key parameters for distinguishing speech from silence (Ramírez; Górriz; Segura, 2007; Yegnanarayana, 1999), contributing to more robust automatic segmentation (Kraljevski; Tan; Bissiri, 2015).

Field (line)	Default	Type	Description
`Save_TextGrid_files (29)`	`1 = true`	boolean	0 = Objects window only; 1 = also save .TextGrid to disk.

5.3 Monitoring progress

The Info window shows 'Realignment process started…' at the beginning and 'Realignment process ended.' with elapsed time at the end.

6. Output TextGrid

The script produces saves a six-tier TextGrid object per audio file. This new TextGrid ([filename].TextGrid) will be used by SpeechRhythmExtractor.

6.1 Complete_ TextGrid tier structure

Tier	Name	Labels	Content
1	`VowelOnsets`	(points)	Point tier — one point per vowel onset
2	`VC`	V, C, #	Phoneme classification: V=vowel, C=consonant, #=pause
3	`ORT-MAU`	orthographic	Orthographic tier from webMAUS
4	`Word`	orthographic	Word-level intervals (orthographic transcription)
5	`Chunk`	CH1, CH2 …	Utterance chunk intervals → Chunk_tier in SpeechRhythmExtractor
6	`Tone`	H1, L1 …	Tone points: H = F0 peak, L = F0 valley (auto or empty)

> *ℹ Tier 3 (MAU/PhonoSyl) retains the original webMAUS phone labels and is the most important tier for manual correction. Tier 5 (V_to_V units) is used by SpeechRhythmExtractor as V_to_V_tier = 1. Tier 2 (VC) is used as V_C_Pause_tier = 2. Tier 9 (Chunk) is used as Chunk_tier = 5 in SpeechRhythmExtractor.*

7. Code walkthrough — line by line

The blocks below document each section of SuppMat_VVunitAligner.praat. Line ranges refer to the actual script (717 lines total).

Lines 1–8 — Header and citation

Lines 1–8 are comment lines (# / ##). They name the script, credit Leônidas Silva Jr. and Florian Schiel's original webMAUS aligner, and provide the citation and repository URL. These lines are never executed by Praat.

Lines 12–30 — FORM — input parameters (form … endform)

Lines 12–30 define the dialog box that appears when the script starts.

- Language (line 17): 1 = Portuguese (BR), 2 = English (US). For BP data, choose Portuguese (BR) — the script will apply the SAMPA-IT phone inventory returned by webMAUS when Italian (IT) is selected.

- Chunk_segmentation (line 23): 1 = Automatic (intensity-based), 2 = Forced/manual (interactive word-list), 3 = None.
- Pause_duration_(s) (line 28): minimum silence gap (seconds) used as a chunk boundary threshold in Automatic mode. Default 0.35 s.
- Save_TextGrid_files (line 29): boolean 0/1. If 1, the final TextGrid is written to disk as filename.TextGrid.

Lines 32–43 — Object cleanup and clearinfo

Lines 32–43 remove any objects from Praat's Objects window before the workflow begins (the if guard prevents a crash on an empty window). clearinfo resets the Info window.

Lines 45–57 — Global variables, file list, progress message, and stopwatch

Lines 45–49 set internal constants: smooth_F0_threshold = 2 (pitch smoothing bandwidth in Hz), window = 0.03, f0step = 0.05 (reserved for future use), spectral_emphasis_threshold = 400.

Line 51 creates a Strings object listing all *.wav files in the current folder; numberOfFiles holds the count for the outer loop.

Lines 53–57 print the 'Realignment process started...' banner to the Praat Info window and start the stopwatch that will report elapsed time at the end.

Lines 59–74 — Outer file loop — loading and naming

Lines 59–74 begin the outer loop (for y from 1 to numberOfFiles). For each .wav file:

- soundname$ = filename with extension; sound_file$ = filename without extension (the Praat Sound object name).
- tg$ = sound_file$ + '.TextGrid' — the webMAUS TextGrid to be loaded.
- maus$ = 'MAUS_' + sound_file$ — name for the preserved original webMAUS copy.
- mausMasVV$ = 'Complete_' + sound_file$ — name for the final output TextGrid.

Lines 65–82 — Load TextGrid and preserve original (MAUS_ copy)

Lines 65–82: the .TextGrid file is loaded (Read from file). The TextGrid is immediately copied and renamed to MAUS_[filename], preserving the original webMAUS segmentation unchanged as one of the three output objects. The working copy is then stripped down to exactly 2 tiers (repeat…until ntiers = 2) using Remove tier: 1, keeping only the MAU phone tier and the ORT syllable tier from the G2P→MAUS→PHO2SYL pipeline.

Lines 84–97 — Create VC and PhonoSyl tiers

Lines 84–87: two new tiers are created by duplication:

- Duplicate tier: 2, 2, 'PhonoSyl' - a copy of the phonological syllable (PHO2SYL) tier at position 3.
- Duplicate tier: 1, 1, 'VC' - a copy of the MAU phone tier at position 1.

Lines 90–97: every interval in the PhonoSyl tier is relabelled. Pause markers <p:> and <p> become '#'; all other non-empty labels become 'PhonoSyl'. This uniform labelling later allows SpeechRhythmExtractor to count phonological syllables directly.

Lines 99–113 — Language dispatch — V/C/# labelling

Lines 99–113: the if/elsif block calls @lang_BP (language = 1) or @lang_AmE (language = 2) to classify each phone interval in the VC tier as V (vowel), C (consonant), or # (pause), based on the SAMPA-IT or X-SAMPA phone inventories respectively.

Lines 113–160 — PROCEDURE lang_AmE and PROCEDURE lang_BP

Procedure lang_AmE (lines 113–135): iterates tier 1 from interval 2 to textVC−1. Each X-SAMPA label returned by the webMAUS English (US) protocol is mapped to V, C, # or '' (glottal stop artefact removal). Vowels include checked vowels (I, E, {, 6, @, U), tense vowels (i:, u:, O:, A:, 3:, 3`), diphthongs (eI, aI, aU, @U, OI…), and the reduced vowel V. Consonants cover all English segments including tS, dZ, liquids, nasals, and the tap (4).

Procedure lang_BP (lines 137–160): same structure but uses the SAMPA-IT inventory returned by webMAUS when Italian (IT) is selected for Brazilian Portuguese data. Vowels cover oral (i, e, eh, a, oh, o, u), nasal (iN, eN, aN, oN, uN), lax variants (I, E, A, O, U), and all diphthongs. Consonants cover the full BP inventory including sh, zh, nh, lh, rr, tS, dZ and the archiphoneme N.

Lines 162–171 — Remove empty-interval boundaries (laryngealization fix)

Lines 162–171: a repair loop iterates tier 1. Whenever an empty interval is encountered (lab$ = ''), its right boundary is removed, merging the empty interval into the adjacent one. This corrects spurious short empty intervals that the MAUS aligner inserts at glottal stop positions. Note that j is NOT incremented when a boundary is removed — the loop re-checks the same position after the merge.

Lines 173–193 — Derive VV (vowel-onset-to-vowel-onset) tier

Lines 173–193:

- Get starting points: 1, 'is equal to', 'V' extracts the onset time of every V interval in tier 1 as a PointProcess ('sound_file$'_V).
- To TextGrid (vuv): 5e-6, 1.25e-6 converts those points to a two-interval TextGrid with very small voiced/unvoiced parameters, placing a boundary at each vowel onset.
- A second repair loop (k = 2, repeat…until) removes all interior left boundaries, collapsing them into a single tier where intervals span consecutive vowel onsets.
- Every internal interval (2 to textVV−1) is labelled 'V_to_V'.

Lines 195–214 — Merge TextGrids and create MAS-VV tier

Lines 195–203: the VC TextGrid and the VV TextGrid are merged with plus / Merge into 'TextGrid merged'. Duplicate tier: 2, 6, 'MAS-VV' creates a copy of the VV tier at position 6; this tier will be trimmed so that each interval starts on the first vowel phone from the webMAUS alignment.

Lines 205–214: the mausMasVV$ copy is renamed; tier 5 receives the name 'V_to_V units'; a VowelOnsets point tier is duplicated from tier 5 and placed at position 1. An ORT-MAU orthographic tier from MAUS is extracted and merged. Chunk (tier 9) and Tone (tier 10) tiers are inserted.

Lines 214–222 — Erase-first dispatch — align MAS-VV to first vowel

Lines 214–222: calls @erase1stMASVVinterval_lang_BP (language = 1) or @erase1stMASVVinterval_lang_AmE (language = 2). Both procedures use a repeat…until loop that removes the left boundary of interval 2 in tier 6 as long as the label is a consonant, stopping when a vowel label is found. This ensures each MAS-VV interval starts on the first vowel, aligning phonological and phonetic syllable boundaries.

Lines 222–280 — Overlap VV and MAS-VV tiers

Lines 275–303: a repeat…until loop (a = 2) compares the duration of each MAS-VV interval (tier 6) to the corresponding VV interval (tier 5). When durPhone < durVV, the right boundary of the MAS-VV interval is removed (extending it). When durPhone ≥ durVV, the counter advances. The loop ends when a = n_intVV, making the MAS-VV intervals co-extensive with the VV units.

Lines 303–326 — Final tier reorganisation

Lines 303–326: the 10-tier merged object is reorganised for SpeechRhythmExtractor compatibility. Three tiers (Chunk, Word, Tone) are duplicated and extracted. The working copy is renamed to sound_file$. The mausMasVV$ (Complete_) object is merged with Word, Chunk, and Tone. Intermediates are removed. The final merged object is renamed to mausMasVV$ (Complete_[filename]).

Lines 326–334 — Chunk segmentation — Mode 2: Forced/manual

When Chunk_segmentation = 2 (lines 326–476): an interactive workflow prompts the user to identify chunk boundaries by word. Capital-letter words are numbered to disambiguate repeated occurrences. A word list is written to disk and displayed. The user specifies the number of chunks, the starting word of each chunk, and the ending word. Boundaries are inserted at the start times of those words in tier 9. Chunks are labelled CH1, CH2 … Tone marks (H/L) are inserted automatically using the Hirst (2012) adaptive pitch floor/ceiling algorithm: a wide-range first-pass Pitch (60–700 Hz) provides Q1/Q3; the second pass uses floor = Q1×0.75 and ceiling = Q3×1.3.

Lines 477–498 — Chunk segmentation — Mode 1: Automatic

When Chunk_segmentation = 1 (lines 477–492): an Intensity object is created (To Intensity... 100 0 0 yes). Boundaries are inserted in tier 9 at every frame with intensity ≥ 40 dB. A cleanup loop then merges all consecutive boundaries. A second pass inspects the VC tier for pause intervals (#) longer than pause_duration seconds and inserts a boundary at each qualifying pause end. Chunks are labelled CH1, CH2 … and Tone marks are added.

Lines 498–512 — Chunk segmentation — Mode 3: None / else branch

When Chunk_segmentation = 3 (the else branch, lines 493–510): an Intensity object is created to detect speech boundaries, the tier is cleaned up, and no chunk labels or Tone points are added. The Chunk and Tone tiers remain empty.

Lines 649–671 — Save TextGrid and remove VV temporary objects

Lines 649–657: if Save_TextGrid_files = 1, @saveTextGrid writes filename.TextGrid to disk.

Lines 654–657: the temporary VV-derived objects (TextGrid 'sound_file$'_V and PointProcess 'sound_file$'_V) are always removed, keeping the Objects window clean.

Lines 673 — End of outer file loop

Line 673: closes the for y (file) loop. The script proceeds to the final summary block.

Lines 676–717 — Final summary, cleanup, and elapsed time

Lines 676–679: a new Strings file list (tgList) counts all .TextGrid files in the folder.

Lines 680–706: if Save_TextGrid_files = 1, the file count is printed and the combined file list is shown in View & Edit. If 0, all Sound and TextGrid objects are removed from the Objects window.

Lines 709–717: the 'Realignment process ended.' banner is printed. The stopwatch is read, split into minutes and rounded seconds, and the elapsed time is reported.

8. Troubleshooting

Symptom	Cause and solution
TextGrid not found	The .TextGrid filename must match the .wav exactly (case-sensitive on macOS/Linux). Ensure webMAUS output is in the script folder.
Script produces no V labels	Wrong language selected, or webMAUS was run with a different pipeline. Verify that G2P→MAUS→PHO2SYL was used.
'TextGrid merged' object error	A previous run left a 'TextGrid merged' object. Close all TextGrid objects and re-run.
Automatic mode produces too many chunks	Reduce Pause_duration_(s) in the form. The default 0.35 s works for most conversational data.
Chunk tier is empty	Chunk_segmentation = 3 (None) was selected, or no silence of the required duration was found.
Save_TextGrid_files = 1 but no file appears	Place the script folder on a short path (Desktop). Long paths can prevent Praat from writing files.

9. Citation

When using this script in publications, please cite:

> *SILVA JR., L. SuppMat_VVunitAligner. [Computer program for Praat], 2021–2026. Available at: https://github.com/leonidasjr/VVunitAlignerCode_webMAUS.*
>
> *KISLER, T.; REICHEL, U. D.; SCHIEL, F. Multilingual processing of speech via web services. Computer Speech & Language, v. 45, p. 326–347, 2017. DOI: https://doi.org/10.1016/j.csl.2016.11.005.*

References (ABNT)

BARBOSA, P. A.; MADUREIRA, S. *Manual de fonética acústica experimental: aplicações a dados do português*. São Paulo: Cortez, 2015.

BISOL, L. *Introdução a estudos de fonologia do português brasileiro*. 5. ed. Porto Alegre: EDIPUCRS, 2010.

BOERSMA, P.; WEENINK, D. Praat: doing phonetics by computer. [Programa de computador], versão 6.3+. Amsterdam: University of Amsterdam, 1992–2026. Available at: https://www.praat.org.

CALLOU, D.; LEITE, Y. *Iniciação à fonética e à fonologia*. 11. ed. Rio de Janeiro: Zahar, 2009.

CANEPARI, L. *Italian phonology*. München: LINCOM Europa, 1999.

KRALJEVSKI, I.; TAN, Z.-H.; BISSIRI, M. P. Comparison of forced-alignment speech recognition and humans for generating reference VAD. In: *Proceedings of INTERSPEECH*, 2015., p. 1–5, 2015.

KISLER, T.; REICHEL, U.; SCHIEL, F. Multilingual processing of speech via web services. *Computer Speech & Language*, v. 45, p. 326–347, 2017. DOI: https://doi.org/10.1016/j.csl.2016.11.005.

LADEFOGED, P. *Phonetic data analysis: an introduction to fieldwork and instrumental techniques*. Malden: Blackwell Publishing, 2003.

LEE, A. Using forced alignment for automatic acoustic-phonetic segmentation. *Proceedings of the International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, p. 1–4, 2012.

MAJOR, R. *Foreign accent: the ontogeny and phylogeny of second language phonology*. Mahwah: Lawrence Erlbaum Associates, 2001.

RAMÍREZ, J; GÓRRIZ, J.; SEGURA, J. Voice activity detection: fundamentals and speech recognition system robustness. In: RAMÍREZ, J. (ed.). *Robust speech recognition and understanding*. Rijeka: InTech, 2007. p. 1–22.

SILVA JR., L.; BARBOSA, P. SpeechRhythmExtractor. [Programa de computador para Praat], 2019–2026. Available at: https://github.com/leonidasjr/SpeechRhythmCode.

YEGNANARAYANA, B. *Speech processing: a dynamic and optimization-oriented approach*. New York: McGraw-Hill, 1999.

YUAN, J.; LAI, W.; CIERI, C.; LIBERMAN, M. Using forced alignment for phonetics research. In: HUANG, Chu-Ren; HSIEH, Shu-Kai; JIN, Pei (eds.). *Chinese language resources*. Springer, 2023. p. 395–414. (Text, Speech and Language Technology, v. 49). DOI: https://doi.org/10.1007/978-3-031-38913-9_17.